My Study Notes

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Contents

1 Introduction

Welcome to my study notes. This document is organized into multiple chapters to keep everything clean and modular.

2 Conjectures and Theorems

2.1 Basic Definitions

Definition 1. *A* conjecture *is a mathematical statement that is believed to be true based on evidence or intuition, but which has not yet been proved or disproved.*

Definition 2. *A* theorem *is a mathematical statement that has been rigorously proved to be true from previously established results (such as axioms, definitions, and other theorems).*

2.2 Examples from Number Theory

Example 2.1 (A true conjecture that became a theorem)**.** *Historically, mathematicians* conjectured *that there are infinitely many prime numbers. This was later proved by Euclid and is now a theorem:*

Theorem 1 (Euclid)**.** *There are infinitely many prime numbers.*

Thus, a statement can start as a conjecture and, once proved, become a theorem.

Example 2.2 (A conjecture that is still open)**.** Goldbach's Conjecture *is one of the most famous unsolved problems in number theory:*

Conjecture 1 (Goldbach)**.** *Every even integer $n \geq 4$ can be written as the sum of two prime numbers.*

This statement has been checked for very large values of n, but no complete proof or counterexample is known.

Example 2.3 (A false conjecture)**.** *Consider the following statement:*

Conjecture 2. *Every odd integer $n > 1$ is a prime number.*

At first glance, small examples like $3, 5, 7$ might suggest this is true. However, it is false. *For instance,*

$$9 = 3 \cdot 3, \quad 15 = 3 \cdot 5,$$

so 9 and 15 are odd but not prime. These are counterexamples that disprove the conjecture.

2.3 Remarks

Remark 1. *Conjectures play a central role in mathematical research. They guide exploration and often lead to new techniques, whether they eventually become theorems or are disproved by counterexamples.*

3 Logical Implication, Necessary Conditions, and Sufficient Conditions

3.1 Introduction

Logic is the language we use to organize mathematical reasoning. Before proving theorems, we need to understand how mathematical statements are connected.

A *statement* is a sentence that is either true or false, but not both.

For example,

"2 is even"

is a true statement, while

"3 is even"

is a false statement.

However,

$$x + 1$$

is not a statement by itself, because it is not true or false. It is only an expression. On the other hand,

$$x + 1 > 0$$

can become a statement once we say what values of (x) we are considering.

In this chapter we introduce implication, necessary conditions, sufficient conditions, logical equivalence, truth tables, and the contrapositive.

3.2 Implication

Let (A) and (B) be statements.

Definition 3.

Implication

The statement

$$A \Rightarrow B$$

means:

if A is true, then B must be true.

We read this as:

"A implies B"

or

"if A, then B."

The implication ($A \Rightarrow B$) does not say that (A) causes (B). It only says that it is impossible for (A) to be true while (B) is false.

So the meaning of

$$A \Rightarrow B$$

is:

there is no case where A is true and B is false.

When (A) is false, the implication makes no demand on (B). For this reason, in classical logic the implication ($A \Rightarrow B$) is considered true whenever (A) is false.

3.3 Necessary and Sufficient Conditions

Implication allows us to describe when one condition guarantees another.

Definition 4.

$$SufficientCondition$$

A statement (A) is a sufficient condition *for (B) if*

$$A \Rightarrow B.$$

This means that (A) being true is enough to guarantee that (B) is true.

In other words, if we know (A), then we automatically know (B).

Definition 5.

$$NecessaryCondition$$

A statement (B) is a necessary condition *for (A) if*

$$A \Rightarrow B.$$

This means that whenever (A) is true, (B) must also be true. Without (B), (A) cannot happen.

Therefore, the same implication

$$A \Rightarrow B$$

has two interpretations:

- (A) is sufficient for (B);
- (B) is necessary for (A).

This is one of the most important ideas in logic.

3.4 Example: Divisibility

Let

$$A = \text{“}n\text{ is divisible by 6”},$$

and

$$B = \text{“}n\text{ is divisible by 3”}.$$

If (n) is divisible by (6), then (n) is divisible by (3). Therefore,

$$A \Rightarrow B.$$

So:

- being divisible by (6) is sufficient for being divisible by (3);
- being divisible by (3) is necessary for being divisible by (6).

However, being divisible by (3) is not sufficient for being divisible by (6). For example,

$$9$$

is divisible by (3), but it is not divisible by (6).
So we have:

$$A \Rightarrow B,$$

but not

$$B \Rightarrow A.$$

3.5 Logical Equivalence

Sometimes two statements imply each other.

Definition 6.

LogicalEquivalence

Statements (A) and (B) are logically equivalent *if*

$$A \Rightarrow B$$

and

$$B \Rightarrow A.$$

In this case we write

$$A \Leftrightarrow B.$$

We read this as:

"A if and only if B."

Logical equivalence means that (A) and (B) always have the same truth value. Either both are true or both are false.

Equivalently,

$$A \Leftrightarrow B$$

means that (A) is both necessary and sufficient for (B), and (B) is both necessary and sufficient for (A).

3.6 Truth Tables

Truth tables allow us to study logical statements by checking all possible truth values.

3.6.1 Implication

The truth table for $(A \Rightarrow B)$ is:

(A)	(B)	$(A \Rightarrow B)$
T	T	T
T	F	F
F	T	T
F	F	T

The only situation where $(A \Rightarrow B)$ is false is when (A) is true and (B) is false.

So the implication

$$A \Rightarrow B$$

is equivalent to saying:

not both A true and B false.

3.6.2 Equivalence

The truth table for $(A \Leftrightarrow B)$ is:

(A)	(B)	$(A \Leftrightarrow B)$
T	T	T
T	F	F
F	T	F
F	F	T

Equivalence is true exactly when (A) and (B) have the same truth value.

3.7 Converse, Inverse, and Contrapositive

Given an implication

$$A \Rightarrow B,$$

we can form three related statements.

Definition 7.

Converse

The converse *of* $(A \Rightarrow B)$ *is*

$$B \Rightarrow A.$$

Definition 8.

Inverse

The inverse *of* $(A \Rightarrow B)$ *is*

$$\neg A \Rightarrow \neg B.$$

Definition 9.

Contrapositive

The contrapositive *of* $(A \Rightarrow B)$ *is*

$$\neg B \Rightarrow \neg A.$$

The converse and inverse are not usually equivalent to the original implication. However, the contrapositive is always equivalent to the original implication.

Theorem 2.

Contrapositive

The implication

$$A \Rightarrow B$$

is logically equivalent to its contrapositive

$$\neg B \Rightarrow \neg A.$$

Proof. The implication $(A \Rightarrow B)$ is false only in one case: when (A) is true and (B) is false.

Now consider the contrapositive:

$$\neg B \Rightarrow \neg A.$$

This implication is false only when $(\neg B)$ is true and $(\neg A)$ is false. But $(\neg B)$ being true means that (B) is false, and $(\neg A)$ being false means that (A) is true.

So the contrapositive is false exactly when (A) is true and (B) is false.

Therefore, $(A \Rightarrow B)$ and $(\neg B \Rightarrow \neg A)$ are false in exactly the same situation. Hence they have the same truth value in every case, so they are logically equivalent. □

3.8 Examples

3.8.1 Geometric Example

Let

$$A = \text{“It is a square”},$$

and

$$B = \text{“It is a rectangle with equal sides”}.$$

Then:

- $(A \Rightarrow B)$: every square is a rectangle with equal sides;
- $(B \Rightarrow A)$: every rectangle with equal sides is a square.

Therefore,

$$A \Leftrightarrow B.$$

So being a square is equivalent to being a rectangle with equal sides.

3.8.2 Number Theory Example

Let

$$A = \text{“}n\text{ is divisible by 6”},$$

and

$$B = \text{“}n\text{ is divisible by 3”}.$$

Then:

$$A \Rightarrow B$$

is true, because every number divisible by (6) is also divisible by (3).
However,

$$B \Rightarrow A$$

is false, because (9) is divisible by (3), but not by (6).
So:

- (A) is sufficient for (B);
- (B) is necessary for (A);
- (A) and (B) are not logically equivalent.

3.8.3 Example with Inequalities

Let

$$A = \text{“}x > 2\text{”},$$

and

$$B = \text{“}x > 0\text{”}.$$

Assume (x) is a real number. If (x¿2), then certainly (x¿0). Therefore,

$$A \Rightarrow B.$$

So (x¿2) is sufficient for (x¿0), and (x¿0) is necessary for (x¿2).
But the converse is not true. For example,

$$x = 1$$

satisfies (x¿0), but does not satisfy (x¿2).
Thus,

$$A \Rightarrow B,$$

but not

$$B \Rightarrow A.$$

3.9 The Importance of the Domain

When working with logical statements in mathematics, we must know the domain of the variables.
For example, the statement

$$x^2 > 0$$

depends on what values of (x) are allowed.
If (x) is a nonzero real number, then

$$x^2 > 0$$

is true.
But if (x=0) is allowed, then the statement is false for (x=0).
Therefore, before deciding whether an implication is true or false, we must know the universe of objects we are discussing.

3.10 Exercises

1. Determine whether the following implication is true or false:
   “‘

   $$\text{“If } n \text{ is prime, then } n \text{ is odd.”}$$

   If it is false, give a counterexample.

2. Determine whether the following implication is true or false:

   $$\text{“If } n \text{ is prime and } n > 2 \text{, then } n \text{ is odd.”}$$

   Identify the sufficient condition and the necessary condition.

3. Let

$$A = \text{“}x > 2\text{”}, \qquad B = \text{“}x > 0\text{”}.$$

   Write the converse, inverse, and contrapositive of $A \Rightarrow B$.

4. Give an example of a condition that is necessary but not sufficient.

5. Give two conditions that are both necessary for a third condition, but neither one alone is sufficient.

6. Construct the truth table for the statement

$$(A \Rightarrow B) \wedge (\neg A \Rightarrow B).$$

   What simpler statement is it logically equivalent to?

7. Let

$$A = \text{“}n \text{ is divisible by } 4\text{”}, \qquad B = \text{“}n \text{ is even”}.$$

   Decide whether $A \Rightarrow B$ is true. Then decide whether $B \Rightarrow A$ is true.

8. Give an example of two statements A and B such that

$$A \Leftrightarrow B.$$

   “‘

4 Algebraic Identities

4.1 Introduction

Algebraic identities are formulas that allow us to rewrite expressions in a more useful or factored form. They play a central role in number theory, especially when studying divisibility, prime numbers, and the structure of integers.

In this chapter we explore classical identities, their applications, and historical curiosities.

4.2 Difference of Powers

One of the most important identities is the factorization of a difference of powers:

Theorem 3 (Difference of Powers)**.** *For any integer x and positive integer n,*

$$x^n - 1 = (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1).$$

This identity is fundamental in the study of numbers of the form $x^n - 1$, especially when $x = 2$.

Example 4.1 (Factoring a Mersenne-Type Number)**.** *Consider the number*

$$2^{15} - 1.$$

Since $15 = 3 \cdot 5$, we can apply the identity twice:

$$2^{15} - 1 = (2^3 - 1)(2^{12} + 2^9 + 2^6 + 2^3 + 1).$$

Then factor $2^3 - 1$ again:

$$2^3 - 1 = (2-1)(2^2 + 2 + 1) = 7.$$

This method reveals the internal structure of the number without guessing prime factors.

4.3 Sum of Powers

Another classical identity is the sum of powers:

Theorem 4 (Sum of Powers)**.** *For any integer x and positive integer n,*

$$x^n + 1 = \begin{cases} (x+1)(x^{n-1} - x^{n-2} + \cdots - x + 1), & \text{if } n \text{ is odd} \\ \\ \text{irreducible over the integers}, & \text{if } n \text{ is even.} \end{cases}$$

Example 4.2 (A Curious Identity)**.** *Since 5 is odd,*

$$2^5 + 1 = 33 = (2+1)(2^4 - 2^3 + 2^2 - 2 + 1).$$

This identity explains why some numbers of the form $2^n + 1$ factor, while others (the Fermat numbers) are suspected to be prime.

4.4 Historical Notes

Remark 2 (Euclid and Mersenne)**.** *Euclid already used the identity*

$$x^n - 1 = (x-1)(x^{n-1} + \cdots + 1)$$

in his proof that there are infinitely many primes. Much later, Marin Mersenne studied numbers of the form $2^p - 1$, now called Mersenne numbers*. Some of these are prime, and they produce the largest known primes in history.*

Remark 3 (Fermat's Curiosity)**.** *Pierre de Fermat conjectured that all numbers of the form*

$$F_n = 2^{2^n} + 1$$

were prime. He was correct for $n = 0, 1, 2, 3, 4$, but Euler disproved the conjecture by showing:

$$F_5 = 2^{32} + 1 = 4294967297 = 641 \cdot 6700417.$$

This is one of the most famous false conjectures in number theory.

4.5 Applications

- Factoring large integers using algebraic structure.
- Understanding prime candidates such as Mersenne and Fermat numbers.
- Simplifying expressions in modular arithmetic.
- Building conjectures about divisibility and primality.

4.6 Curiosities

- The largest known prime numbers are almost always of the form $2^p - 1$, where p is prime.
- The identity for $x^n - 1$ is used in cryptography, especially in algorithms involving cyclic groups.
- Some identities, like the sum of powers, behave differently depending on whether the exponent is odd or even — a subtle but important detail.

5 Limits, Infinitesimals, Continuity, and the Derivative

In arithmetic, we enlarged the number system whenever an equation could not be solved.
In calculus, a different kind of difficulty appears.
Sometimes we want to understand a value that is not reached directly, but is approached.
For example, consider the numbers

$$2, \quad 1.5, \quad 1.333\ldots, \quad 1.25, \quad 1.2, \quad \ldots$$

They seem to move closer and closer to (1), but no term in this list is exactly (1).
This raises a new question.

How can a variable have a final tendency even if it never reaches that value?

To study this question, consider the sequence

$$x_n = 1 + \frac{1}{n}.$$

When (n=1),

$$x_1 = 2.$$

When (n=2),

$$x_2 = \frac{3}{2}.$$

When (n=10),

$$x_{10} = 1.1.$$

When (n=1000),

$$x_{1000} = 1.001.$$

The values are not equal to (1), but they become closer and closer to (1).
So the important object is not equality.
The important object is distance.
The distance from

$$(x_n)$$

to (1) is

$$|x_n - 1|.$$

Computing,

$$|x_n - 1| = \left|1 + \frac{1}{n} - 1\right| \frac{1}{n}.$$

Now we ask:

Can this distance be made smaller than any positive number chosen in advance?

Suppose someone gives us a small number ($\varepsilon > 0$).
We want

$$|x_n - 1| < \varepsilon.$$

Since

$$|x_n - 1| = \frac{1}{n},$$

we need

$$\frac{1}{n} < \varepsilon.$$

This is true whenever

$$n > \frac{1}{\varepsilon}.$$

Therefore, no matter how small (ε) is, we can choose (n) large enough so that all later values of (x_n) are within distance (ε) of (1).

This phenomenon deserves a name.

Definition 10. *A number (a) is called the* limit *of a sequence (x_n) if, for every ($\varepsilon > 0$), there exists an integer (N) such that whenever (n>N),*

$$|x_n - a| < \varepsilon.$$

In this case we write

$$\lim_{n\to\infty} x_n = a.$$

Thus,

$$\lim_{n\to\infty}\left(1 + \frac{1}{n}\right) = 1.$$

5.1 Problems

Problem 1

Consider

$$x_n = \frac{1}{n}.$$

What number does (x_n) approach as ($n \to \infty$)?

Problem 2

Show that

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

That is, given ($\varepsilon > 0$), find a condition on (n) which guarantees

$$\left|\frac{1}{n} - 0\right| < \varepsilon.$$

Problem 3

Consider

$$x_n = 3 + \frac{2}{n}.$$

What is the limit?

Prove your answer using the definition.

6 The Limit of a Function

Sequences are not the only objects that approach values.

A function can also approach a value.

Consider

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

At first, something strange happens when (x=1).

If we substitute (x=1), we get

$$f(1) = \frac{1^2 - 1}{1 - 1} = \frac{0}{0},$$

which is not defined.

So the function has no value at (x=1).

But what happens near (x=1)?

Since

$$x^2 - 1 = (x - 1)(x + 1),$$

we have, for ($x \neq 1$),

$$f(x) = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

Therefore, when (x) is close to (1), the values of (f(x)) are close to

$$1 + 1 = 2.$$

For example,

$$f(0.9) = 1.9,$$

$$f(0.99) = 1.99,$$

$$f(1.001) = 2.001.$$

Even though (f(1)) is not defined, the nearby values of (f(x)) clearly approach (2).

This creates a new question.

> Can a function approach a value at a point even if the function is not defined at that point?

The answer is yes.

This is why we need the limit of a function.

Definition 11. *Let (f(x)) be a function. We say that*

$$\lim_{x \to a} f(x) = b$$

if, for every ($\varepsilon > 0$), there exists a ($\delta > 0$) such that whenever

$$0 < |x - a| < \delta,$$

we have

$$|f(x) - b| < \varepsilon.$$

The condition

$$0 < |x - a|$$

means that (x) is allowed to approach (a), but (x) is not required to equal (a).
This is essential.
A limit is about nearby behavior.
It is not necessarily about the value at the point.

6.1 Example

Let

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

For ($x \neq 1$),

$$f(x) = x + 1.$$

Therefore,

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \lim_{x \to 1} (x + 1)2.$$

So the function does not have a value at (x=1), but it has a limit there.

6.2 Problems

Problem 4

Compute

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}.$$

First simplify the expression for ($x \neq 2$).

Problem 5

Consider

$$f(x) = \begin{cases} x + 1, & x \neq 1, \\ 10, & x = 1. \end{cases}$$

What is

$$\lim_{x \to 1} f(x)?$$

What is (f(1))?
Are they equal?

7 Infinitesimals

When (x) approaches (a), the difference

$$x - a$$

approaches (0).
When (f(x)) approaches (b), the difference

$$f(x) - b$$

also approaches (0).
So many limit questions reduce to studying quantities that become smaller and smaller.
This suggests another name.

Definition 12. *A function* $(\alpha(x))$ *is called an* infinitesimal *as* $(x \to a)$ *if*

$$\lim_{x \to a} \alpha(x) = 0.$$

Similarly, $(\alpha(x))$ *is called an infinitesimal as* $(x \to \infty)$ *if*

$$\lim_{x \to \infty} \alpha(x) = 0.$$

For example,

$$\frac{1}{x}$$

is infinitesimal as $(x \to \infty)$, because

$$\lim_{x \to \infty} \frac{1}{x} = 0.$$

Also,

$$x - a$$

is infinitesimal as $(x \to a)$, because

$$\lim_{x \to a} (x - a) = 0.$$

7.1 A Limit Written as a Constant Plus an Error

Suppose a variable (y) approaches (b).
Then the difference

$$y - b$$

approaches (0).
So (y-b) is infinitesimal.
Let

$$\alpha = y - b.$$

Then

$$y = b + \alpha.$$

This means that a quantity approaches (b) exactly when it can be written as

$$\text{constant} + \text{infinitesimal}.$$

Theorem 1

A variable (y) satisfies

$$\lim y = b$$

if and only if it can be written in the form

$$y = b + \alpha,$$

where (α) is infinitesimal.

Proof. If

$$y = b + \alpha$$

and ($\alpha \to 0$), then

$$y - b = \alpha.$$

Therefore,

$$|y - b| = |\alpha|.$$

Since (α) is infinitesimal, ($|\alpha|$) can be made smaller than any ($\varepsilon > 0$). Hence

$$y \to b.$$

Conversely, if ($y \to b$), define

$$\alpha = y - b.$$

Then

$$\alpha \to 0,$$

so (α) is infinitesimal, and

$$y = b + \alpha.$$

7.2 Algebra of Infinitesimals

Once infinitesimals appear, we must ask whether they behave well under ordinary operations.

> If two quantities become negligible, does their sum also become negligible?

Theorem 2

The sum of finitely many infinitesimals is infinitesimal.

Proof. Let

$$u(x) = \alpha(x) + \beta(x),$$

where both (α) and (β) are infinitesimal.
Given ($\varepsilon > 0$), choose (x) close enough to the limiting value so that

$$|\alpha(x)| < \frac{\varepsilon}{2}$$

and

$$|\beta(x)| < \frac{\varepsilon}{2}.$$

Then

$$|u(x)| = |\alpha(x) + \beta(x)| \leq |\alpha(x)| + |\beta(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus (u(x)) is infinitesimal.

Theorem 3

The product of an infinitesimal and a bounded function is infinitesimal.

Proof. Let ($\alpha(x)$) be infinitesimal and let (z(x)) be bounded.
Since (z(x)) is bounded, there exists (M¿0) such that

$$|z(x)| \leq M.$$

Given ($\varepsilon > 0$), choose (x) close enough so that

$$|\alpha(x)| < \frac{\varepsilon}{M}.$$

Then

$$|z(x)\alpha(x)| = |z(x)|, |\alpha(x)| \leq M|\alpha(x)| < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

Therefore ($z(x)\alpha(x)$) is infinitesimal.

Theorem 4

If ($\alpha(x)$) is infinitesimal and ($\alpha(x) \neq 0$), then

$$\left|\frac{1}{\alpha(x)}\right|$$

becomes arbitrarily large.

Proof. Let (M¿0). Since ($\alpha(x) \to 0$), we can choose (x) close enough so that

$$|\alpha(x)| < \frac{1}{M}.$$

Then

$$\left|\frac{1}{\alpha(x)}\right| = \frac{1}{|\alpha(x)|} > M.$$

So the reciprocal of a nonzero infinitesimal becomes arbitrarily large.

8 Continuity

We have seen that a function may have a limit at a point even if it is not defined there.

We have also seen that a function may be defined at a point but have a value different from the nearby behavior.

For example,

$$f(x) = \begin{cases} x+1, & x \neq 1, \\ 10, & x = 1. \end{cases}$$

Then

$$\lim_{x \to 1} f(x) = 2,$$

but

$$f(1) = 10.$$

So the function has a break at (x=1).

This raises another question.

> When does the value of a function agree with the value approached by nearby points?

This is the idea of continuity.

Definition 13. *A function (f(x)) is called* continuous *at* (x_0) *if*

$$\lim_{x \to x_0} f(x) = f(x_0).$$

In words:

> A function is continuous at (x_0) when the value it approaches is the value it actually has.

8.1 Continuity Written with Increments

Let

$$x = x_0 + \Delta x.$$

Then

$$\Delta x = x - x_0.$$

As $(x \to x_0)$, we have

$$\Delta x \to 0.$$

The change in the function is

$$\Delta y = f(x_0 + \Delta x) - f(x_0).$$

If (f) is continuous at (x_0), then as $(\Delta x \to 0)$,

$$\Delta y \to 0.$$

Therefore, continuity can also be written as

$$\lim_{\Delta x \to 0} \big(f(x_0 + \Delta x) - f(x_0)\big) = 0.$$

This says:

> A small change in (x) produces a small change in (y).

8.2 Direct Substitution

If (f) is continuous at (x_0), then

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Since

$$x_0 = \lim_{x \to x_0} x,$$

we may write informally

$$\lim_{x \to x_0} f(x) = f\left(\lim_{x \to x_0} x\right).$$

This is why continuous functions allow limits to be computed by direct substitution.

8.3 Problems

Problem 6

Let

$$f(x) = x^2.$$

Show that (f) is continuous at ($x_0 = 3$).
That is, show that

$$\lim_{x \to 3} x^2 = 9.$$

Problem 7

Let

$$f(x) = \begin{cases} x^2, & x \neq 2, \\ 7, & x = 2. \end{cases}$$

Find

$$\lim_{x \to 2} f(x).$$

Find (f(2)).
Is (f) continuous at (2)?

9 Continuity on an Interval

Continuity at one point is a local property.
But sometimes we want a function to behave without breaks over a whole interval.

Definition 14. *A function (f(x)) is continuous on an interval ((a,b)) if it is continuous at every point of that interval.*

If the interval is closed,

$$[a, b]$$

,

then (f) must be continuous at every interior point, continuous from the right at (a), and continuous from the left at (b).

10 Properties of Continuous Functions

Once continuity is understood as "no breaks," we can ask what such functions are forced to do.
For example, suppose a function is continuous on a closed interval

$$[a, b].$$

Can it grow forever inside this interval?
No.
The interval is finite, and the function has no breaks that allow it to escape.
This leads to an important theorem.

Theorem 5 (Extreme Value Theorem)

If (f(x)) is continuous on a closed interval ([a,b]), then there exist points ($x_1, x_2 \in$ [a,b]) such that

$$f(x_1) \geq f(x)$$

and

$$f(x_2) \leq f(x)$$

for every ($x \in [a, b]$).
In other words, a continuous function on a closed interval attains a maximum and a minimum.

11 A Problem Before the Derivative: Fermat's Segment Problem

Before defining the derivative, let us begin with a problem.
Suppose we have a segment (AB) of fixed length (L).
Choose a point (C) between (A) and (B).
Let

$$AC = x.$$

Then

$$CB = L - x.$$

We want to choose (C) so that the product

$$AC \cdot CB$$

is as large as possible.
In symbols, we want to maximize

$$P(x) = x(L - x).$$

11.1 First Experiments

If (C) is very close to (A), then (AC) is small.
So

$$AC \cdot CB$$

is small.
If (C) is very close to (B), then (CB) is small.
Again,

$$AC \cdot CB$$

is small.
So the maximum should happen somewhere in the middle.
Let us test some values.
If

$$x = 0,$$

then

$$P(0) = 0(L - 0) = 0.$$

If

$$x = \frac{L}{4},$$

then

$$P\left(\frac{L}{4}\right) = \frac{L}{4}\left(L - \frac{L}{4}\right)\frac{L}{4} \cdot \frac{3L}{4}\frac{3L^2}{16}.$$

If

$$x = \frac{L}{2},$$

then

$$P\left(\frac{L}{2}\right) = \frac{L}{2}\left(L - \frac{L}{2}\right)\frac{L}{2} \cdot \frac{L}{2}\frac{L^2}{4}.$$

The middle seems better.
But experiments are not proof.
We need a method.

11.2 Moving the Point Slightly

Suppose the point (C) is already chosen.
Now move it a very small distance (h) to the right.
Then the new value of (AC) is

$$x + h.$$

The new value of (CB) is

$$L - x - h.$$

So the new product is

$$P(x + h) = (x + h)(L - x - h).$$

The old product was

$$P(x) = x(L - x).$$

The question becomes:

Did the product increase or decrease after this small movement?

To answer this, compute the difference

$$P(x+h) - P(x).$$

We have

$$P(x+h) - P(x) = (x+h)(L-x-h) - x(L-x).$$

Expand the first product:

$$(x+h)(L-x-h) = x(L-x-h) + h(L-x-h).$$

So

$$(x+h)(L-x-h) = x(L-x) - xh + h(L-x) - h^2.$$

Therefore,

$$P(x+h) - P(x) = x(L-x) - xh + h(L-x) - h^2 - x(L-x).$$

Canceling (x(L-x)), we get

$$P(x+h) - P(x) = -xh + h(L-x) - h^2.$$

Thus,

$$P(x+h) - P(x) = h(L-2x) - h^2.$$

11.3 The Important Part of the Change

The change in the product is

$$P(x+h) - P(x) = h(L-2x) - h^2.$$

Now notice something important.
If (h) is very small, then (h^2) is much smaller than (h).
For example, if

$$h = 0.01,$$

then

$$h^2 = 0.0001.$$

So the term

$$h(L-2x)$$

is the main part of the change.
The term

$$-h^2$$

is smaller.
This suggests that the expression

$$L - 2x$$

controls whether the product first wants to increase or decrease.

11.4 The Balance Condition

At the maximum, moving slightly to the right should not improve the product.

Also, moving slightly to the left should not improve the product.

So at the best point, there should be no first-order advantage in either direction.

That means the main part of the change must vanish:

$$L - 2x = 0.$$

Solving,

$$2x = L.$$

Hence

$$x = \frac{L}{2}.$$

Therefore,

$$AC = \frac{L}{2}$$

and

$$CB = \frac{L}{2}.$$

So the product ($AC \cdot CB$) is largest when (C) is the midpoint of (AB).

11.5 What Was Discovered

We did not begin with the derivative.

We began with a concrete problem.

Then we asked what happens when the input changes slightly.

This forced us to compare

$$P(x+h)$$

with

$$P(x).$$

The difference

$$P(x+h) - P(x)$$

measured the change in the quantity.

Then we found that the most important part of the change was the part proportional to (h):

$$h(L - 2x).$$

At the maximum, this first-order part had to disappear.

This is the beginning of the derivative.

> The derivative is born from the attempt to understand which part of a small change controls increase and decrease.

11.6 Challenge

Try the same idea with

$$P(x) = x^2.$$

Compute

$$P(x + h) - P(x).$$

Which term is proportional to (h)?
Which term is proportional to (h^2)?
What expression controls the first-order change?

11.7 Challenge

Now try

$$P(x) = x^3.$$

Compute

$$P(x + h) - P(x).$$

Expand:

$$(x + h)^3 - x^3.$$

Which part is proportional to (h)?
What expression appears multiplying (h)?
This expression will later become the derivative of (x^3).

12 The Need for a Derivative

Continuity tells us when a function has no breaks.
But continuity does not tell us how fast the function changes.
Consider a moving object.
At time (x), suppose its position is

$$y = f(x).$$

If time changes from (x) to $(x + \Delta x)$, then the position changes from

$$f(x)$$

to

$$f(x + \Delta x).$$

So the change in position is

$$\Delta y = f(x + \Delta x) - f(x).$$

The average rate of change is

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

This is the slope of the secant line between the two points

$$(x, f(x))$$

and

$$(x + \Delta x, f(x + \Delta x)).$$

But now we ask a sharper question.

What is the rate of change at one instant?

An instant has no length, so we cannot simply divide by zero.

Instead, we take smaller and smaller intervals and look for the value approached by the average rates of change.

This creates the derivative.

Definition 15. *Let (y=f(x)). For an increment* $(\Delta x \neq 0)$*, define*

$$\Delta y = f(x + \Delta x) - f(x).$$

If the limit

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

exists, then this limit is called the derivative *of (f) at (x), and is denoted by*

$$f'(x)$$

or

$$\frac{dy}{dx}.$$

Thus,

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

12.1 Example: The Derivative of (x^2)

Let

$$f(x) = x^2.$$

Then

$$f(x + \Delta x) = (x + \Delta x)^2.$$

So

$$\Delta y = f(x + \Delta x) - f(x)(x + \Delta x)^2 - x^2.$$

Expanding,

$$(x + \Delta x)^2 = x^2 + 2x\Delta x + (\Delta x)^2.$$

Therefore,

$$\Delta y = x^2 + 2x\Delta x + (\Delta x)^2 - x^2 2x\Delta x + (\Delta x)^2.$$

Now divide by (Δx):

$$\frac{\Delta y}{\Delta x} = \frac{2x\Delta x + (\Delta x)^2}{\Delta x} 2x + \Delta x.$$

Taking the limit as ($\Delta x \to 0$),

$$f'(x) = \lim_{\Delta x \to 0} (2x + \Delta x) 2x.$$

Thus,

$$\frac{d}{dx}(x^2) = 2x.$$

12.2 What We Have Discovered

We began with quantities that approach values without necessarily reaching them.

This forced us to define limits.

Then we noticed that many differences approach zero.

This forced us to name infinitesimals.

Then we asked when the approached value of a function equals the actual value.

This forced us to define continuity.

Finally, we asked how to measure instantaneous change.

This forced us to define the derivative.

> The derivative is not a formula first. It is the answer to the problem of measuring change at an instant.

12.3 Problems

Problem 8

Using the definition of derivative, compute the derivative of

$$f(x) = x.$$

Problem 9

Using the definition of derivative, compute the derivative of

$$f(x) = x^3.$$

Problem 10

Let

$$f(x) = |x|.$$

Compute the average rate of change

$$\frac{f(0 + \Delta x) - f(0)}{\Delta x}.$$

What happens when ($\Delta x > 0$)?

What happens when ($\Delta x < 0$)?

Does the derivative exist at (x=0)?

13 Algebraic Number Theory

13.1 Why New Numbers Appear

The natural numbers are sufficient to solve equations such as

$$x + 3 = 5.$$

However, they are not sufficient to solve

$$x + 5 = 3.$$

This leads to the introduction of negative numbers.
Likewise, integers are not sufficient to solve

$$2x = 1,$$

which motivates the rational numbers.
The rational numbers are not sufficient to solve

$$x^2 = 2,$$

which motivates the real numbers.
The real numbers are not sufficient to solve

$$x^2 + 1 = 0,$$

which motivates the complex numbers.
A recurring pattern appears throughout mathematics:

> When an equation has no solution in the current number system, we enlarge the system.

The study of numbers is therefore inseparable from the study of equations.

14 A New Number

Consider the equation

$$x^2 + 5 = 0.$$

Introduce a symbol (u) satisfying

$$u^2 = -5.$$

We denote this new number system by

$$\mathbb{Z}[u] = \{a + bu : a, b \in \mathbb{Z}\}.$$

Since $u^2 = -5$, we may also write

$$\mathbb{Z}[\sqrt{-5}].$$

We now allow expressions of the form

$$a + bu,$$

where $(a, b \in \mathbb{Z})$.
Examples are

$$3 + u, \qquad 7 - 2u, \qquad 5.$$

This creates a new arithmetic world containing the integers together with a solution of

$$x^2 + 5 = 0.$$

15 An Unexpected Observation

In the integers, the number

$$6$$

has the factorization

$$6 = 2 \cdot 3.$$

Using ($u^2 = -5$), we compute

$$(1 - u)(1 + u) = 1 - u^2 = 1 - (-5) = 6.$$

Therefore

$$6 = (1 - u)(1 + u).$$

At first sight this appears harmless.
Is this really a different factorization of 6?
The next problems investigate this question.
This raises a fundamental question:

> Can arithmetic behave differently in different number systems?

The remainder of this book is an attempt to understand this phenomenon.

15.1 Problem 1

Compute:

$$(1 + u)(1 - u)$$

Use the fact:

$$u^2 = -5$$

What ordinary integer do you obtain?

15.2 Problem 2

Compute:

$$(2 + u)(2 - u)$$

15.3 Problem 3

Look the number:

$$6 = 3.2$$

Now compare with your answers from problems 1 and 2.

15.4 Problem 4

Can 2 divide 1+u such that:

$$1 + u = 2(a + bu)$$

?

15.5 Problem 5

Can 3 divide 1+u such that:

$$1 + u = 3(a + bu)$$

?

Surprisingly, none of the factors appears to divide the others.

This suggests that

$$6 = 2 \cdot 3 = (1 - u)(1 + u)$$

may genuinely be two different factorizations of the same number.

If this is true, then unique factorization has failed.

16 Norm

The factorization

$$(a + bu)(a - bu) = a^2 + 5b^2$$

produces an ordinary integer.

This suggests associating to each number $a + bu$ the integer

$$N(a + bu) = a^2 + 5b^2,$$

a ordinary integer, that we will call *Norm* and use the symbol N to compute it. The norm allows us to measure elements of $\mathbb{Z}[u]$ using ordinary integers.

Since

$$N(a + bu) = (a + bu)(a - bu),$$

many questions about factorization in $\mathbb{Z}[u]$ can be translated into questions about ordinary integers.

16.1 Problem 6

Prove that

$$N(\alpha\beta) = N(\alpha)N(\beta).$$

This property is called multiplicativity of the norm.

16.2 Problem 7

Compute the norms:

$$\begin{aligned} N(1 + u) \\ N(2) \\ N(3) \end{aligned}$$

Now suppose

$$1 + u = \alpha\beta.$$

Using the multiplicativity of the norm,

$$N(1 + u) = N(\alpha)N(\beta).$$

Since

$$N(1+u) = 1^2 + 5(1)^2 = 6,$$

we obtain

$$6 = N(\alpha)N(\beta).$$

The positive factorizations of 6 are

$$1 \cdot 6$$

and

$$2 \cdot 3.$$

Can any element of $\mathbb{Z}[u]$ have norm 2?
Can any element of $\mathbb{Z}[u]$ have norm 3?
If not, the only possibility is

$$N(\alpha) = 1$$

or

$$N(\beta) = 1.$$

A norm equal to 1 corresponds to a unit.
Therefore one factor must be a unit, and $1+u$ is irreducible.

We said irreducible not prime.

16.3 Problem 8

In ordinary integers, if p is prime and

$$p|ab$$

then

$$p|a$$

or

$$p|b$$

Now test this here with:

$$3|(1+u)(1-u)$$

Why does 3 divide the product but not the either factor?

Answering this makes you start understanding why we do not use the prime word rather irreducible.

1.What should irreducible mean?
2.What should prime mean?

17 irreducible and Primes

17.1 irreducibles

From our observations we can define that a number r is irreducible if whenever

$$r = ab$$

one of the factors is a unit, 1 or -1. In the world of a+bu the only units are 1 and -1.

Thus 1+u is irreducible because every factorization has unit factor.

17.2 Primes

Now for prime using our new world $\mathbb{Z}[\sqrt{-5}]$ we already discover that 3 fails because it is irreducible but not prime.

Remember that a number p is prime if whenever:

$$p|ab$$

then:

$$p|a$$

or

$$p|b$$

Notice that this definition does not mention factorization of p itself. It is a divisibilty property.

Now historically in ordinary integers:

irreducible = prime

But in our new world $\mathbb{Z}[\sqrt{-5}]$ they are different.

17.3 Problem 9

Can you prove that 1+u, 1-u and 2 are irreducible?

Hint: Use the norm N.

18 Where We Are

We began by enlarging the integers in order to solve an equation.

The new numbers behaved naturally under addition and multiplication.

Yet something unexpected occurred: a familiar theorem about unique factorization appeared to fail.

The number

$$6 = 2 \cdot 3 = (1 - u)(1 + u)$$

seems to admit two genuinely different factorizations.

The remainder of this book investigates how arithmetic changes when the number system changes.

What should we do next?

- Redefine primes?
- Redefine factorization?
- Search for a new object that factors uniquely even when numbers do not?

The third possibility turns out to be the most fruitful.
After all, the numbers themselves seem perfectly reasonable.
Addition works.
Multiplication works.
The norm works.
What appears to have failed is factorization.
This suggests a new question.

Is there a hidden object behind numbers that still factors uniquely?

Let us return to the mysterious identity

$$6 = 2 \cdot 3 = (1+u)(1-u).$$

Imagine there were objects

$$A, \quad B, \quad C, \quad D$$

such that

$$2 = AB,$$

$$3 = CD,$$

and

$$1 + u = AC,$$

$$1 - u = BD.$$

Then we would obtain

$$6 = 2 \cdot 3 = (AB)(CD) = ABCD$$

and also

$$6 = (1+u)(1-u) = (AC)(BD) = ABCD.$$

Suddenly the factorization becomes unique again.
We do not know whether such objects actually exist.
We only know what properties they would need to have.
This gives us a clue.

Perhaps numbers are only shadows of a finer factorization system.

The next step is to investigate divisibility more carefully.

19 Divisibility Sets

Consider all numbers divisible by (2).
In ordinary integers this gives

$$\ldots, -4, -2, 0, 2, 4, \ldots.$$

What does the analogous set look like in $\mathbb{Z}[\sqrt{-5}]$?

19.1 Problem 10

Assume a+bu is divisible by 2.

Then

$$a + bu = 2(c + du).$$

What conditions must (a) and (b) satisfy?

19.2 Problem 11

Describe explicitly the set of all numbers divisible by 2.

Call this set $\{2\}$,

19.3 Problem 12

Suppose

$$x, y \in \{2\}.$$

Show that

$$x + y \in \{2\}.$$

In other words, the sum of two elements divisible by 2 is again divisible by 2.

19.4 Problem 13

Suppose

$$x \in \{2\}$$

and

$$r \in \mathbb{Z}[\sqrt{-5}].$$

Show that

$$rx \in \{2\}.$$

Thus multiplying by any element of the $\mathbb{Z}[\sqrt{-5}]$ keeps us inside the set.

We have discovered a remarkable kind of subset.

It is closed under addition.

It is also stable under multiplication by arbitrary elements of the $\mathbb{Z}[\sqrt{-5}]$.

Such subsets appear naturally whenever we study divisibility.

Because they occur so frequently, they deserve a name.

20 Ideals

An *ideal* is a subset I of ($\mathbb{Z}[\sqrt{-5}]$) satisfying:

1. If x,y∈ I, then x+y∈ I.
2. If x∈ I and r∈ $\mathbb{Z}[\sqrt{-5}]$, then rx∈I.

The set:

$$\{2\} = \{2(a + bu) : a, b \in \mathbb{Z}\}$$

is our first example.
Likewise we may consider

$$\{3\},$$

the set of all numbers divisible by 3,
and

$$\{(1 + u)\},$$

the set of all numbers divisible by (1+u).
At first sight these sets seem to contain less information than the numbers themselves.
Surprisingly, the opposite is true.
As we shall discover, ideals remember factorization more faithfully than numbers do.

21 Prime Ideals

Now to save factorization again we start thinking what would be a prime ideal? At the beginning of mathematics, numbers seemed fundamental but after enough algebraic number theory, one starts thinking:

> The polynomial is primary and the numbers are shadows of the polynomial.

For example:

$$x^2 + 1$$

creates the Gaussian integers.

$$x^2 + 5$$

creates $\mathbb{Z}[\sqrt{-5}]$.

$$x^2 - 2$$

creates $\mathbb{Z}[\sqrt{2}]$.
Different polynomials create a different arithmetic universes.

21.1 Problem 14

Suppose α satisfies

$$\alpha^2 - 2 = 0$$

abd β satifies

$$\beta^4 - 4\beta^2 + 2 = 0$$

How could you tell whether β lives in the same arithmetic world as α?

I hope you did a break and try to solve it because now you are in the edge of a great discover. Treat y = x^2:

$$y^2 - 4y + 2$$

Solving Baskara we find:

$$y = \frac{4 \pm \sqrt{16 - 8}}{2} = 2 \pm \sqrt{2}$$

Therefore the roots satisfy:

$$x^2 + \sqrt{2}$$

or

$$x^2 - \sqrt{2}$$

Now, notice something remarkable. The polynomial contains only integers:

$$x^4 - 4x^2 + 2$$

Yet, it's roots invole $\sqrt{2}$, so:

> The world generated by it's roots seems connected to the world generated by $\sqrt{2}$.

This makes us think:

> Does a root of one polynomial already live inside the arithmetic universe generated by another root?

or

> When do two different polynomials generate the same arithmetic world?